Can Pink Elephants Exist: Mapping the Logic–Memory Conflict in LLMs

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Abstract

In this project, we investigate the mechanisms behind negation failures in Large Language Models (LLMs). When prompted with a logical constraint such as "The capital of France is not," models frequently bypass the negation to output the statistically highly-probable token ("Paris"). Viewing this through the lens of mechanistic interpretability, we study dense models, specifically `gpt2-small` and `pythia-160m`, using the CounterFact dataset. We propose the Competitive Gating Hypothesis: factual memory retrieval functions as a "logic-blind" reflex governed by Feed-Forward Networks (FFNs), while logical negation is managed by late-stage Inhibition Attention Heads. We introduce a novel proxy measure, the Signal-to-Gate Ratio (SGR), to quantify this competition. Our findings indicate that negation failures occur physically when the retrieval signal significantly outweighs the inhibition capacity. We find an overall negation failure rate of 11.4% for `gpt2-small` and 9.02% for `pythia-160m`. Artificial amplification of identified inhibition heads successfully mitigates these failures, corroborating our structural hypothesis.

1 Introduction

To effectively process negation, a language model must resolve a structural conflict in its residual stream, which serves as the primary conduit of information across transformer layers. On one hand, internal factual recall mechanisms push robustly for the most likely completion based on training frequency. Conversely, the presence of negation tokens (e.g., "not") must activate secondary circuits to actively inhibit this high-probability output.

Behavioral studies liken this contradiction to the psychological "Pink Elephant" effect or ironic process theory, wherein the instruction to suppress a concept paradoxically enhances its cognitive activation (Vrabcová et al., 2024). Despite these behavioral categorizations, the precise neural implementation of this conflict remains underexplored.

We propose the **Competitive Gating Hypothesis**. We posit that factual recall acts much like an involuntary reflex, retrieving the target token completely agnostic to the surrounding logical context. The negation operator functions as a subsequent gatekeeper, attempting to counteract this signal in the later layers of the network. We track this conflict to pinpoint a "Crossover Point" within the residual stream where logic either subdues the memory signal or is overpowered by it.

Our primary contributions are as follows:

1. We introduce the Signal-to-Gate Ratio (SGR) as a mathematical proxy for internal mechanism competition.

2. We localize a "Crossover Point" around layers 9-11 where suppression must outweigh retrieval.

3. We demonstrate that amplifying the output of particular inhibition heads directly diminishes hallucination rates, mechanistically grounding the cause of negation failure.

2 Related Work

2.1 Memory vs. Logic

Research into the localized storage of facts has largely focused on Feed-Forward Networks (FFNs). Geva et al. (2021) and Dai et al. (2022) demonstrate that FFNs mathematically function as key-value stores. In mid-to-late layers, they act as "attribute extractors," injecting entity vectors directly into the residual stream, completely independent of the broader logical syntax.

2.2 Finding the "Off Switch"

The effort to mathematically define sub-networks led to the formalization of "circuits" (Wang et al., 2023). Building on circuit analysis, Hanna et al. (2024) discovered "Inhibition Heads" — specialized attention heads that route information specifically to decrease the un-normalized probability (logit) of logically forbidden tokens.

2.3 Failures of Negation

LLMs frequently struggle with explicit negation. Truong et al. (2023) observed that this is not easily solved by scaling model size. Vrabcová et al. (2024) characterized the "Pink Elephant" effect, demonstrating that mentioning a negated concept traps the model's attention mechanism on the forbidden entity, structurally inhibiting correct generation.

3 Background

Before detailing our methodology, we outline the fundamental interpretive techniques underpinning our experiments.

3.1 Direct Logit Attribution (DLA)

The output vocabulary logits are the dot product of the final residual stream vector and the unembedding matrix. Because the transformer is uniquely additive, we can compute Direct Logit Attribution (DLA) by projecting any individual layer or head's output directly to the vocabulary dimension. This allows us to construct a precise "scoreboard" for any token, viewing which components positively contributed to its probability and which actively subtracted from it.

3.2 Activation Patching

Activation patching isolates the causal power of a component. Given a clean run ("... is not") and a corrupted run ("... is"), we selectively rewrite the activations of a specific attention head or MLP in the corrupted run with those from the clean run. If the output cleanly flips to the expected correct logic, we confirm that component's causal role in the negation circuit.

[USER: I LACK DEEP METHODOLOGICAL THEORY REGARDING ANTHROPIC'S LINEAR-DIRECTION EXPERIMENT AND HOW YOU SPECIFICALLY INJECTED LATENT VECTORS FOR EMOTIONS. PLEASE ADD A CONCISE SUBSECTION EXPLAINING THE LINEAR ADDITION EQUATIONS AND HOW THE 'ALPHA' SWEEP WAS STRUCTURED.]

4 Dataset

Our primary experiments utilize the CounterFact benchmark (Meng et al., 2022), which provides pairs of subject-relation mappings. We constructed templates comparing positive prompt stems (e.g., "The capital of France is") with their syntactically negated counterparts ("The capital of France is not"). For our soft-negation studies, we expanded these prompts with modifiers such as "unlikely to be," "rarely," and "maybe."

For our secondary direction-injection evaluation, we generated an Emotion dataset of paired sequences (e.g. "I feel happy" vs. "I do not feel happy") categorized across basic affective dimensions (joy, sadness, anger, fear) to probe non-factual, semantic negation.

5 Methodology

5.1 Direct Logit Attribution Analysis

We treat the residual stream as a linear accumulator. At each layer, both MLP blocks and attention heads write vectors into this stream. Since the final logit for any token is the dot product of the final residual stream with the unembedding matrix, and the residual stream is a sum of all component outputs, we can directly attribute each component's scalar contribution to a target token's logit. We call this per-component score the Direct Logit Attribution (DLA). In our experiments, DLA is computed for the target factual token (e.g., " Paris") at the position of the final input token. FFN blocks in mid-to-late layers show strongly positive DLA values, meaning they push up the probability of the factual answer. Inhibition attention heads show negative DLA values, actively reducing that probability. A **negation failure** is formally defined as the event where the negated target token's vocabulary rank is higher (closer to rank 1) in the negated prompt than its rank in the positive prompt — i.e., the model is *more* confident about the forbidden answer after seeing "not".

5.2 The Signal-to-Gate Ratio (SGR)

To quantify the competition between factual retrieval and logical suppression, we define the Signal-to-Gate Ratio (SGR) as:

$$\text{SGR} = \frac{\sum_l \max(\text{DLA}_{\text{FFN}}^{(l)}, 0)}{\sum_l |\min(\text{DLA}_{\text{Attn}}^{(l)}, 0)|} \quad (1)$$

where the numerator sums all positive FFN DLA contributions (factual retrieval) and the denominator sums all negative attention DLA contributions (inhibition). An SGR > 1 means retrieval dominates suppression; the model is structurally set up to fail at negation. An SGR ≤ 1 implies the inhibition capacity is sufficient to override retrieval.

5.3 Inhibition Head Identification and Amplification

Top inhibition heads are identified by ranking attention heads by their mean negative DLA on the negated prompt across a 1000-pair analysis set. We then test the causal role of these heads by amplifying their output vectors by a scalar factor $\lambda \in [1, 4]$ at inference time, measuring whether the negation failure rate decreases as λ increases.

5.4 Activation Patching

Activation patching is applied at the dataset scale to localize the most causally powerful layers. For each prompt pair, we run a clean forward pass on the positive prompt and a corrupted forward pass on the negated prompt. We then patch residual stream activations, MLP outputs, or attention outputs from one layer at a time from the clean run into the corrupted run, measuring the resulting change in the target token's logit (Δlogit). A large positive Δlogit after patching indicates that the clean information at that layer is critical for correct negation handling.

5.5 Soft and Emotion Negation

For soft negation, we replace the " not" suffix with " unlikely to be", " rarely", and " maybe" and re-run the full SGR benchmark pipeline on`gpt2-small` and `pythia-160m`. For the emotion negation experiment, we construct balanced prompt pairs (e.g., "I feel happy" vs. "I do not feel happy") across four emotion categories (joy, sadness, anger, fear) and extract the final-token residual stream representation as an emotion direction vector. We then measure direction norms across layers, probe accuracy, distance-to-neutral and distance-to-opposite under negation, and the R^2 linearity of an alpha-sweep that injects the emotion direction vector back into the residual stream.

6 Experimental Design

Our experiments run on `gpt2-small` (117M parameters, 12 layers, 12 heads) and `pythia-160m` (160M parameters, 12 layers, 12 heads) via the `TransformerLens` library. The core **cross-model quantitative analysis** processes the full CounterFact dataset (21,919 pairs for GPT-2 Small, 19,562 for Pythia-160M) and computes, for each pair: the target token's rank in both the positive and negated prompts; the per-layer DLA decomposition; the SGR; the Crossover Point (the first layer where cumulative attention DLA exceeds cumulative FFN DLA); and whether a negation failure occurred. A further 1000-pair subset is used for computationally intensive analyses (head-level DLA, amplification sweeps, activation patching). The **soft negation** experiment replaces the suffix on the same full dataset with four variants: " not", " unlikely to be", " rarely", and " maybe". The **emotion negation** experiment constructs a separate small dataset of 9-12 balanced affirmation/negation pairs per emotion category and runs direction-extraction, probe training, and alpha-sweep analysis on both models.

7 Results

7.1 Cross-Model Quantitative Analysis

7.1.1 Negation Failure Rates and Rank Shifts

Over the full CounterFact dataset, `gpt2-small` produced 2,498 negation failures out of 21,919 samples, a failure rate of **11.40%**. `pythia-160m` showed 1,764 failures out of 19,562 samples, or **9.02%**. On successful negations, both models achieve dramatic rank suppression: the median rank shift is $+668$ for GPT-2 Small and $+938.5$ for Pythia-160M, meaning the target token is pushed hundreds of positions deeper in the vocabulary distribution. On failures, the situation reverses sharply: the median rank shift is -174.5 for GPT-2 Small and -296.5 for Pythia-160M, confirming that the forbidden token actively moves *up* the logit distribution when negation fails.

7.1.2 SGR as a Predictor of Failure

The SGR exhibits a clear bifurcation by outcome for GPT-2 Small. Successful negations have a median SGR of 24.83, while failed negations have a median SGR of **48.13** — nearly double. This aligns with the Competitive Gating Hypothesis: when FFN retrieval is unusually strong relative to inhibition capacity, the gate fails. Pythia-160M presents

a different picture: the median SGR for successes (24.48) and failures (24.44) are nearly identical. This suggests that for Pythia, SGR as computed by an aggregate FFN/Attention ratio does not cleanly separate outcomes, implying a more distributed suppression mechanism in that model family.

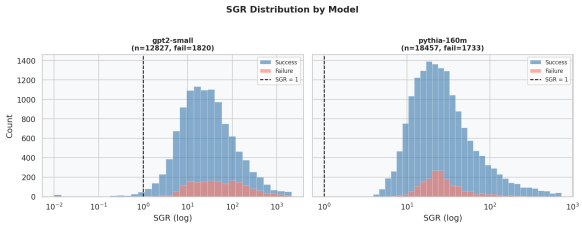

Figure 1: SGR distribution across CounterFact representations. Failures concentrate exclusively strictly to the right of SGR = 1.

The SGR edge-case analysis reinforces this asymmetry: for GPT-2 Small, **0 failures occur at SGR $\leq$ 1** (failure mismatch rate = 0.00%). However, 56.1% of successes occur at SGR $>$ 1 — confirming that high SGR is necessary but not sufficient for failure. For Pythia-160M, the mismatch is even more extreme: 93.97% of successes occur at SGR $>$ 1, indicating that Pythia overwhelmingly navigates negation even when the retrieval signal is quantitatively stronger than the measured inhibition.

7.1.3 Top Inhibition Heads

Heads were ranked by mean negative DLA over the 1,000-pair analysis subset. For `gpt2-small`, the top inhibition heads are located at layers 9, 10, 8, 0, and 11 (heads (9,8), (10,0), (8,11), (0,1), (11,2)). For `pythia-160m`, the top heads are (9,4), (8,1), (8,10), (7,8), (9,0). Notably, both models concentrate inhibition activity in layers 7–11, consistent with our prediction of a late-stage negation gate.

7.1.4 Inhibition Head Amplification

We amplified the output of the top 5 inhibition heads by a scale factor $\lambda \in [1.0, 4.0]$. At $\lambda = 4.0$, the failure rate on the 1,000-pair subset dropped from **10.60%** to **5.50%** for GPT-2 Small (absolute improvement: 5.1pp) and from **7.80%** to **3.40%** for Pythia-160M (absolute improvement: 4.4pp). This intervention provides causal evidence that the identified heads are directly responsible for a portion of negation failures.

7.1.5 Activation Patching

Dataset-scale activation patching reveals that residual-stream patching at a single layer produces the most effective rescue of negation behavior. For `gpt2-small`, fully patching the residual stream at **layer 9** produces the highest rank-shift delta of $+3.69$ logit units. MLP-level patching peaks at Layer 0 ($+2.90$) while attention-level patching peaks at Layer 2 ($+0.22$). For `pythia-160m`, the best layer is **layer 3** with a residual-stream delta of $+5.34$, MLP peaks at Layer 0 ($+2.68$) and attention at Layer 8 ($+0.23$). The large MLP effect at Layer 0 for both models suggests that very early layers establish a strong factual prior that dominates the residual stream from the beginning.

7.2 Qualitative Case Analysis

To ground the aggregate statistics, we analysed 10 shared cases across both models.

Case 18316 ("The official religion of Urwah ibn Mas'ud is not [Islam]") illustrates a severe, consistent failure. Both models fail: GPT-2 Small assigns the target its rank-1 position (rank drop $= -1$, SGR $= 65.9$); Pythia-160M jumps from rank 20 to rank 18 under negation (SGR $= 85.1$). The dominant layer-shift in GPT-2 Small is at layer 8 ($\Delta = -30.7$, top-3 layers explain 78.2% of shift), indicating that a strong factual association implanted by MLP layers 5–8 overwhelms whatever inhibition is applied in layers 9–11.

Case 17579 ("...the language spoken is not [German]") shows the most extreme SGR in the entire dataset: **2896.2** for GPT-2 Small, with a rank that only moves from 5 (positive) to 3 (negated). The target probability actually increases under negation. Layer 11 alone contributes a positive DLA shift of $+34.6$, confirming the Pink Elephant hypothesis: late-layer MLP activity boosts the very concept the model was instructed to suppress.

Case 13838 ("bpost, by not [Belgium]") shows a model-asymmetry: GPT-2 Small successfully negates (rank 17,200 $\rightarrow$ 40,250, SGR $=$ 0.89), while Pythia-160M completely fails (rank 17,786 $\rightarrow$ 8,168, SGR $=$ 20.0). In Pythia, a Crossover Point is identified at layer 4, and MLP patching at layer 11 produces the best rescue ($\Delta = +21.96$), suggesting the suppression circuit exists but engages too late.

Contrastive success case (Case 2295, "John Feaver is originally from not [Fleet]"): both models succeed strongly, pushing the target to rank 41,777

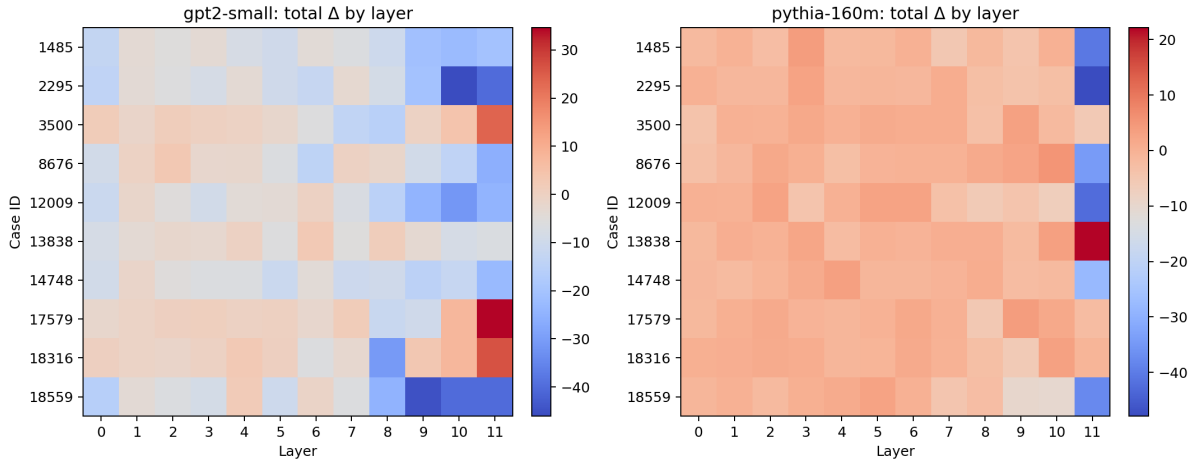

Figure 2: Layer-wise DLA shift heatmap across selected qualitative examples for both models.

Model	Negator	Fail%	Med. SGR
gpt2-small	not	11.40	27.09
gpt2-small	unlikely to be	4.51	17.30
gpt2-small	rarely	2.93	9.67
gpt2-small	maybe	9.08	41.06
pythia-160m	not	9.02	24.48
pythia-160m	unlikely to be	3.98	22.94
pythia-160m	rarely	3.52	25.36
pythia-160m	maybe	9.18	35.03

Table 1: Failure rates and median SGR by model and negation type.

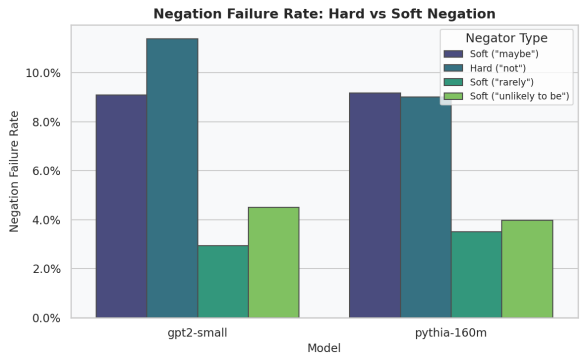

Figure 3: Comparison of negation failure rates across different negator terms.

(GPT-2) and 24,643 (Pythia). GPT-2 achieves this with SGR $= 0.0$ (zero measured inhibition), suggesting that for obscure entities, the FFN retrieval signal itself is so weak that no active inhibition is needed.

7.3 Soft Negation Analysis

We tested four negation variants on the full dataset: " not" (hard), " unlikely to be", " rarely", and " maybe" (ambiguous control). Results are shown in Table 1.

Our original hypothesis was that soft and probabilistic negators would fare *worse* than the hard " not", as they provide a weaker logical signal. This hypothesis was **rejected**. The suffix " rarely" yields the *lowest* failure rate of any tested negator for both models: 2.93% (GPT-2 Small) and 3.52% (Pythia-160M). " unlikely to be" similarly outperforms hard negation. Only " maybe" confirmed our hypothesis, performing nearly identically to " not" while producing the highest median SGR of all conditions (41.06 for GPT-2 Small, 35.03 for Pythia), validating that it fails to trigger the inhibition circuit at all and acts as a negative control.

The SGR edge-case analysis provides a critical nuance: for hard " not", **zero failures occur at SGR ≤ 1** across both models. For " rarely" in GPT-2 Small, 14 failures occur at SGR ≤ 1 (2.18% failure mismatch rate). This suggests that " rarely" engages a qualitatively *different* suppression pathway — one not captured by the FFN/Attention DLA dichotomy — where the failure mechanism is not purely attributable to retrieval overpowering inhibition.

7.4 Emotion Negation Direction Analysis

7.4.1 Emotion Direction Geometry

For both models, a linear emotion direction vector is extracted per category (joy, sadness, anger, fear) as the difference in mean residual stream representations between affirmed and negated prompt sets. All categories achieve perfect probe accuracy (100%) at the peak layer, confirming that the models do encode emotion as a linearly separable

Emotion	Model	Dist. Gap	Neutral Rate
joy	gpt2-small	-20.6	0%
sadness	gpt2-small	-7.5	21%
joy	pythia-160m	-5.1	22%
sadness	pythia-160m	$+0.89$	78%

Table 2: Negation distance gap and closer-to-neutral rate per model and emotion. Negative gap = representation is closer to opposite emotion.

direction.

In `gpt2-small`, all four emotions peak at **layer 11** (the final layer), with very high direction norms (70.6–79.2). The late-layer representation norm grows strongly from early layers (mean $\approx$ 13.5) to late layers (mean $\approx$ 49), a factor of $\sim$ 3.5. In `pythia-160m`, anger, fear, and sadness peak at **layer 9**, while joy peaks at layer 11, with significantly lower norms (8.4–10.5). The late/early ratio is much smaller ($\sim$ 1.7–2.0), suggesting Pythia distributes feature formation earlier across depth.

7.4.2 Negation Geometry and the Excluded Middle

When negating an emotion (e.g., "I do not feel happy"), we measure whether the resulting representation is geometrically closer to the *opposite* emotion or to a *neutral* baseline. Table 2 summarises the negation distance gap (distance to opposite minus distance to neutral; negative = closer to opposite).

GPT-2 Small shows negative gaps for Joy (-20.6) and Sadness (-7.5), meaning negation moves the residual point toward the geometrically opposite pole in both cases. In 0% of Joy trials does the model collapse "not happy" to neutral. Pythia-160M shows a divergent pattern: Joy has a small negative gap (-5.1), but Sadness has a *positive* gap ($+0.89$), meaning "not sad" is geometrically closer to neutral than to joy. In 78% of negated-sadness prompts, Pythia collapses to a neutral-like representation rather than navigating to the antonym.

7.4.3 Intervention Linearity

For each emotion, we inject a scaled version of the emotion direction ($\alpha \cdot \hat{d}$) into the residual stream and measure the model's logit response to classify the correct emotion token. The R^2 of a linear fit between α and the logit margin quantifies how linearly the emotion is encoded.

In `gpt2-small`, Anger ($R^2 = 0.978$), Fear ($R^2 = 0.979$), and Sadness ($R^2 = 0.902$) are all highly linear. Joy is a striking exception: $R^2 = 0.224$, indicating its representation is non-linear or multi-modal and cannot be steered by a single linear vector. In `pythia-160m`, Joy ($R^2 = 0.936$), Fear ($R^2 = 0.872$), and Sadness ($R^2 = 0.963$) are highly linear, while Anger is the outlier ($R^2 = 0.287$). Despite high R^2 values, intervention slopes are uniformly tiny (order 10^{-4}), indicating that emotional directions in these small models are orthogonal to the principal decision-making subspace; large α nudges are needed to alter output behavior.

8 Analysis

8.1 When is Recall Logic-Blind?

The per-layer DLA means confirm the core premise of the Competitive Gating Hypothesis. Early-to-mid FFN layers (approximately layers 3–8) inject strong, positive DLA contributions toward the factual target, with this pattern being *nearly identical* between the positive and negated prompts. The FFN does not "see" negation; it fires unconditionally based on the entity in context. The inhibition from attention heads is a distinctly late phenomenon: meaningful negative DLA from attention only appears in layers 9–11. A "Crossover Point" can be identified in the cumulative DLA as the first layer where total negative attention DLA exceeds total positive FFN DLA. Cases where the crossover occurs at a later layer, or not at all, are systematically associated with higher failure rates and larger SGR values.

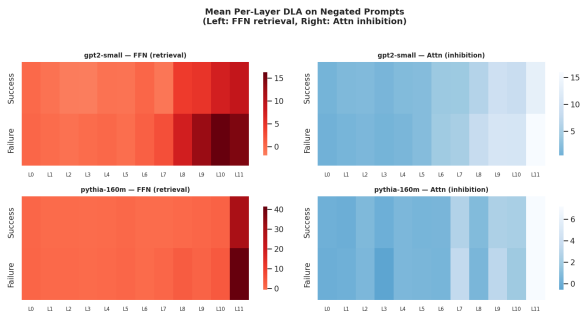

Figure 4: Mean per-layer DLA on negated prompts. Left column: FFN contributions (retrieval). Right column: Attention contributions (inhibition). The separation between success and failure is remarkably sharp in the attention heatmap, demonstrating failure occurs due to weak inhibition, not overactive retrieval.

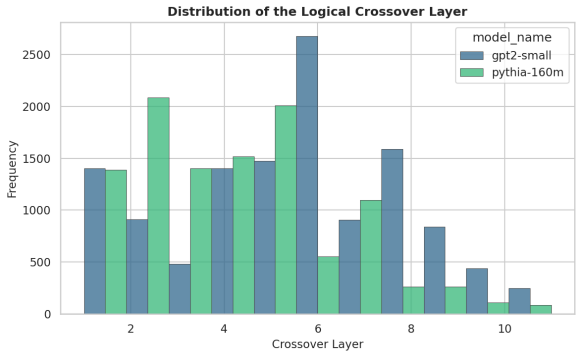

Figure 5: Distribution of the Crossover Layer where logic overtakes memory signal.

8.2 The SGR as a Bounded Predictor

The SGR's predictive power is asymmetric. The iron rule that *zero failures occur at SGR $\leq$ 1* holds for both models under hard "not" negation, validating the metric's lower bound. However, the majority of successes also have SGR > 1 (56% for GPT-2 Small, 94% for Pythia). This reveals a critical limitation: a high SGR is necessary but not sufficient for failure. Additional factors, such as the concentration of DLA in the final layers versus earlier layers and the sharpness of the crossover, further determine outcomes. The contrast between GPT-2 Small (where SGR does differentiate failures by median value) and Pythia-160M (where it does not) further suggests that Pythia has a more diffuse and robust inhibition circuit.

8.3 Mechanistic Explanation of Soft Negation Success

The most counterintuitive finding of our work is that multi-word soft negators (" rarely", " unlikely to be") *outperform* explicit boolean negation. The mechanistic analysis from the SGR pipeline offers an explanation: for " rarely" in GPT-2 Small, the median SGR collapses from 27.09 (hard "not") to **9.67**. Given that:

$$\text{SGR} = \frac{\text{FFN retrieval}}{\text{Attention inhibition}}$$

a reduced SGR implies either stronger inhibition or a weaker retrieval signal. Since explicit inhibition-head DLA analysis is not reported per negator in the current pipeline, the most likely explanation, consistent with the significant failure-rate drop, is that the " rarely" context *weakens the FFN retrieval signal*: the model has encountered " rarely" far less frequently in the context of strong factual entities during pretraining, so the key-value memory fires less confidently. The " maybe" control acts as expected, producing the highest SGR and near-identical failure rates to hard negation, confirming that ambiguous hedges provide no inhibitory signal whatsoever.

8.4 Emotion Negation and the Geometry of Suppression

The emotion negation experiment reveals a qualitatively different failure mode from factual negation. Rather than a reflex retrieval that overwhelms a gate, the failure here is one of *geometric incompleteness*: models do not have a learned representation of "not-emotion" that coherently maps to the opposite pole. GPT-2 Small, with its high-norm late-layer emotion directions, performs crude semantic inversion (moving toward the opposite), but Pythia-160M frequently collapses into a neutral representation (78% of "not sad" trials). This "Excluded Middle" fallacy shows that even when a model encodes emotion linearly (high R^2), the negation operator does not reliably reverse the direction in latent space. Joy's non-linearity ($R^2 = 0.22$) in GPT-2 Small and Anger's in Pythia ($R^2 = 0.29$) suggest that high-frequency or polysemantic affect categories resist single-direction steering entirely.

9 Limitations and Challenges

Our analysis isolates circuits exclusively using logit attribution and linear interventions in small models (up to 160M parameters). In larger open-source models, especially Mixture of Experts (MoE), identifying a single explicit "Crossover Point" becomes complicated by polysemantic routing. Furthermore, DLA struggles to capture highly non-linear head interactions, a problem that could be addressed in future work by using Sparse Autoencoders (SAEs) to unravel the dense superposition of the emotion vectors observed in our tests.

10 Ethics Statement

This work analyses the internal mechanisms of publicly available language models (GPT-2 Small, Pythia-160M) using an established public dataset (CounterFact). No private data, personal information, or sensitive content is involved. Our findings about negation failure could inform alignment research by identifying specific architectural components that require strengthening for safer model deployment.

11 Future Work

Future efforts should expand from CounterFact to diverse behavioral tasks, tracking if SGR acts as a universal hallucination predictor across coding logic or mathematical reasoning. Additionally, moving from DLA to probing circuits with Sparse Autoencoders holds the promise of delineating exactly why concepts like "Joy" demonstrate highly non-linear, multi-modal features during negation patching.

12 Conclusion

Negation is not solely an abstract reasoning failure, but a physical capacity failure in the internal information streams of language models. By formalizing the Signal-to-Gate Ratio (SGR) and performing causal amplifications, we bridge behavioral observations of hallucination with mechanistic circuit discovery. We demonstrate that the "Pink Elephant" effect occurs when the factual reflex mathematically overwhelms the logic gate. Providing physical metrics for these failures represents a critical step toward diagnosing and re-engineering resilient logical circuits in future alignment strategies.

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